Minor Project (21CSA391A)

Proposal (Individual Mode)

Title: Chatbot for mental health support using NLP
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(applicable for the students opting for industry integration)

Address of the Company:

Name, Designation and Communication details of the Guide:

Abstract

This project focuses on building an intelligent mental health support chatbot that understands user context, emotions, and intent using deep learning-based Natural Language Processing (NLP) models. Unlike rule-based systems, the chatbot utilizes transformer-based models such as BERT, DistilBERT, or GPT for contextual understanding and sequence-to-sequence models for generating responses.

The system analyzes user messages to detect mental health indicators such as anxiety, depression, and stress, and provides personalized supportive responses. It leverages datasets such as the Kaggle Mental Health Conversational Dataset, Reddit Mental Health Support Chat Data, and Google's GoEmotions dataset for training and evaluation.

The chatbot is implemented using Python with frameworks like TensorFlow or PyTorch and Hugging Face Transformers, and deployed using Flask. The system aims to provide accessible and scalable emotional support through an AI-driven conversational interface.

Assumptions/Declarations:

The chatbot provides **supportive responses only** and does not replace professional medical or psychological advice.
It is assumed that users interact in **English language text format**.
The project relies on **public datasets (Kaggle, Reddit, GoEmotions)**, which may contain bias or incomplete emotional representation.
Transformer models (BERT, GPT) are assumed to provide sufficient contextual understanding for accurate predictions.
Internet and computational resources are assumed to be available for model training and deployment.
User data privacy and ethical considerations will be maintained throughout the project.

Main Objective/Deliverable:

(Outlines the primary goal or purpose of the project and identifies the specific outcomes or products that the project aims to produce.)

Objective:

To develop an AI-based chatbot capable of understanding user emotions and intent and generating meaningful, context-aware supportive responses using NLP techniques.

Deliverables:

- Emotion detection model (BERT / RoBERTa / LSTM-based)
- Intent recognition module (CNN / Transformer encoder)
- Response generation system (Seq2Seq or GPT-based)
- Working chatbot interface (Flask / Streamlit web application)
- Integration of datasets and trained models
- Storage system for chat history (MongoDB / SQLite)

Timeline and Milestones:

	Milestones	Timeline
1.	Requirement analysis and dataset collection	WEEK 1
2	Data preprocessing and cleaning	WEEK 2
3	Emotion detection model implementation	WEEK 3
4	Intent recognition module development	WEEK 4
5	Response generation model integration	WEEK 5
6	Backend development using Flask/Streamlit	WEEK 6
7	Frontend chatbot interface	WEEK 7
8	Testing and debugging	WEEK 8
9	Final deployment and documentation	WEEK 9

Tools to be used.

Software/Hardware Tools	Specifications
Programming Language: Python **Libraries/Frameworks:** • TensorFlow / PyTorch • Hugging Face Transformers **Datasets:** • Kaggle Mental Health Conversational Dataset • Reddit Mental Health Support Chat Data • GoEmotions (Google – 58 emotion labels) **Web Development:** Flask / Streamlit **Database:** MongoDB / SQLite	

Learning involved:

Topic	Description
Natural Language Processing (NLP) fundamentals Transformer models (BERT, DistilBERT, GPT) Emotion detection and intent classification Sequence-to-sequence models for response generation Model training and evaluation using deep learning frameworks Web application development (Flask / Streamlit) Integration of AI models into real-world applications Ethical considerations in AI-based mental health systems	

Date
Student Name and Signature